Measured pipeline traces

Five-Stage Pipelined Microcontroller for a Washing Machine
Logisim-Evolution 5.0.0 • Assignment 2

Recorded from the submitted gate-level circuit using the unmodified Logisim Evolution 5.0.0 simulator. Each row is settled immediately before a rising edge; writes listed commit on that edge. Reset edge is cycle 0. IF shows the current fetch PC, including cycles where fetch is frozen/killed. ID/EX/MEM/WB show the instruction PC only when their valid bit is set. Full raw logs and test ROMs are in the separately supplied testing package.

PC addresses and byte values in the traces are hexadecimal; cycle numbers are decimal. V = valid bits in WB,MEM,EX,ID order. FwdA/B: R=register/ID bypass value, E=EX/MEM, W=MEM/WB. S=stall, F=flush (branch or HALT). Invalid entries are shown as –. A HALT flush at the end of each test is expected.

1 8.2 - exactly one load-use bubble

PC 04 is LD R1,[R2], PC 05 is ADD R3,R1. R2=0x10, RAM[0x10]=07, R3=05. Expected ADD result=0C and one stall.

Cycle	IF	ID	EX	MEM	WB	V	S/F	A/B	Architectural write
0	00	–	–	–	–	0000	0/0	R/R	-
1	00	–	–	–	–	0000	0/0	R/R	-
2	01	00	–	–	–	0001	0/0	R/R	-
3	02	01	00	–	–	0011	0/0	R/R	-
4	03	02	01	00	–	0111	0/0	R/R	-
5	04	03	02	01	00	1111	0/0	E/W	R2=10
6	05	04	03	02	01	1111	0/0	R/R	R0=07; [10]=07
7	06	05	04	03	02	1111	1/0	R/R	-
8	06	05	–	04	03	1101	0/0	W/R	R3=05
9	07	06	05	–	04	1011	0/0	R/W	R1=07
10	08	07	06	05	–	0111	0/1	R/R	-
11	08	–	–	06	05	1100	0/0	R/R	R3=0C
12	08	–	–	–	06	1000	0/0	R/R	-
13	08	–	–	–	–	0000	0/0	R/R	-
14	08	–	–	–	–	0000	0/0	R/R	-
15	08	–	–	–	–	0000	0/0	R/R	-
16	08	–	–	–	–	0000	0/0	R/R	-

No duplicate store: EX/MEM always advances during a younger stall; only valid EX/MEM ST produces STORE=1. The recorded store pulse count agrees with the expected transaction count. See validation.pdf for assertions and observed values.

2 8.3 - youngest producer forwards to store

PC 04 ADD R1,R2 yields 0D; PC 05 SUB R1,R3 yields 09; PC 06 ST R1,[R4]. R4=0x12. Expected one store RAM[12]=09, not stale 0D.

Cycle	IF	ID	EX	MEM	WB	V	S/F	A/B	Architectural write
0	00	–	–	–	–	0000	0/0	R/R	-
1	00	–	–	–	–	0000	0/0	R/R	-
2	01	00	–	–	–	0001	0/0	R/R	-
3	02	01	00	–	–	0011	0/0	R/R	-
4	03	02	01	00	–	0111	0/0	R/R	-
5	04	03	02	01	00	1111	0/0	R/R	R1=0A
6	05	04	03	02	01	1111	0/0	R/R	R2=03
7	06	05	04	03	02	1111	0/0	R/R	R3=04
8	07	06	05	04	03	1111	0/0	E/R	R4=12
9	08	07	06	05	04	1111	0/0	E/R	R1=0D
10	09	08	07	06	05	1111	0/0	R/R	R1=09; [12]=09
11	0A	09	08	07	06	1111	0/1	R/R	-
12	0A	–	–	08	07	1100	0/0	R/R	R5=09
13	0A	–	–	–	08	1000	0/0	R/R	-
14	0A	–	–	–	–	0000	0/0	R/R	-
15	0A	–	–	–	–	0000	0/0	R/R	-
16	0A	–	–	–	–	0000	0/0	R/R	-

No duplicate store: EX/MEM always advances during a younger stall; only valid EX/MEM ST produces STORE=1. The recorded store pulse count agrees with the expected transaction count. See validation.pdf for assertions and observed values.

3 8.4 - branch kills wrong-path store and ADD

PC 03 SUB R1,R1 makes zero. PC 04 BEQZ targets 07. PC 05 ST and PC 06 ADD must be invalidated. Target LDI R3,55 executes. A separate 8_4_flush test replaces wrong-path ADD with HALT and also passes.

Cycle	IF	ID	EX	MEM	WB	V	S/F	A/B	Architectural write
0	00	–	–	–	–	0000	0/0	R/R	-
1	00	–	–	–	–	0000	0/0	R/R	-
2	01	00	–	–	–	0001	0/0	R/R	-
3	02	01	00	–	–	0011	0/0	R/R	-
4	03	02	01	00	–	0111	0/0	R/R	-
5	04	03	02	01	00	1111	0/0	R/R	R1=05
6	05	04	03	02	01	1111	0/0	R/R	R2=01
7	06	05	04	03	02	1111	0/1	E/R	R4=F4
8	07	–	–	04	03	1100	0/0	R/R	R1=00
9	08	07	–	–	04	1001	0/0	R/R	-
10	09	08	07	–	–	0011	0/0	R/R	-
11	0A	09	08	07	–	0111	0/1	R/R	-
12	0A	–	–	08	07	1100	0/0	R/R	R3=55
13	0A	–	–	–	08	1000	0/0	R/R	-
14	0A	–	–	–	–	0000	0/0	R/R	-
15	0A	–	–	–	–	0000	0/0	R/R	-
16	0A	–	–	–	–	0000	0/0	R/R	-

No duplicate store: EX/MEM always advances during a younger stall; only valid EX/MEM ST produces STORE=1. The recorded store pulse count agrees with the expected transaction count. See validation.pdf for assertions and observed values.